

# MATH503 Lecture

Tuesday  
8:00 to 9:50  
WA220

# MATH503 Mathematics for Computing

## 1.5 Conditional probability

# Housekeeping!

- ▶ Tutorial solutions available from Fridays at 4pm
- ▶ MyiMaths logins were handed out in Week 1
- ▶ First portfolio group activity was done last week
- ▶ Your first quiz is open.

Conditional probability

Independent events

Product rule of probability

Product rule and tree diagrams

Product rule for independent events

Monty Hall problem

Conditional Probability helps Data Scientists to get better results from the given data set.



For Machine Learning Engineers, it helps in building more accurate models for predictions.



The probability of selling a TV on a given normal day may be only 30%.



But if we consider that the given day is a public holiday, then the probability of selling a TV is higher.

The conditional probability of selling a TV on a day given that day is a public holiday might be 70%.

We can represent these probabilities as:

$P(\text{TV sale, given that it is a random day}) = 30\%$ .

$P(\text{TV sale, given that it is a public holiday}) = 70\%$ .

TV sale = TV; Random day = R; Public holiday = H

$P(\text{TV} | \text{R}) = 30\%$ .

$P(\text{TV} | \text{H}) = 70\%$ .

# Conditional probability

The **conditional probability** of event  $E$ , given event  $F$  is

$$P(E|F) = \frac{P(E \cap F)}{P(F)} \quad \text{where } P(F) \neq 0.$$

## Example 1 (manual pp 18)

There are 200 students in a school of which 80 are female students. Of the 200 students, 140 are studying astronomy of which 40 are female. Let  $F$  be the event that a student is female and  $A$  the event that a student studies astronomy.

(a) Sketch a Venn diagram for the problem.

# Conditional probability

(b) Complete the following table:

|                        | Female | Male | Total |
|------------------------|--------|------|-------|
| Study astronomy        |        |      |       |
| Do not study astronomy |        |      |       |
| Total                  |        |      | 200   |

(c) What is the probability that a student is studying astronomy?

$$P(A) =$$



# Conditional probability

(d) If the student is female, what is the probability that she is studying astronomy?  
Use the Venn diagram or the table to calculate this probability.

$$P(A|F) =$$

(e) Use the formula for conditional probability to find the probability that the student studies astronomy, given that the student is female.

$$P(A|F) = \frac{P(A \cap F)}{P(F)}$$

$$P(F) =$$

$F \cap A$  or  $A \cap F$ : student studies astronomy and is female

$$P(A \cap F) =$$

$$P(A|F) =$$

# Conditional probability

(f) Find  $P(A|F^I)$  and explain the meaning of this conditional probability.

From the table:

$$P(A \cap F^I) =$$

$$P(F^I) =$$

$$P(A|F^I) = \frac{P(A \cap F^I)}{P(F^I)} =$$

|                        | Female | Male | Total |
|------------------------|--------|------|-------|
| Study astronomy        | 40     | 100  | 140   |
| Do not study astronomy | 40     | 20   | 60    |
| Total                  | 80     | 120  | 200   |

$P(A|F^I)$  is the probability that a student studies astronomy, given that the student is not a woman

# Independent events

Events  $E$  and  $F$  are **independent events** if

$$P(F|E) = P(F) \quad \text{or} \quad P(E|F) = P(E)$$

## Example 2

Are the events  $F$  and  $A$  in Example 1 independent events?

# Product rule of probability

If  $E$  and  $F$  are events, then  $P(E \cap F)$  may be found by a product formula.

$$P(E \cap F) = P(F)P(E|F) \quad \text{or} \quad P(E \cap F) = P(E)P(F|E).$$

# Product rule of probability

## Example 3

A box contains 6 good items and 2 defective items. Two items are selected at random from the box without replacement. Find the probability that both selected items are good.

## Solution

# Product rule of probability

## Example 3

A box contains 6 good items and 2 defective items. Two items are selected at random from the box without replacement. Find the probability that both selected items are good.

## Solution

$E$ : the item selected first is good

$F$ : the item selected second is good

We have to find  $P(E \cap F)$ .

$P(E) = \frac{6}{8}$ . Explain why.

What is the sample space  $S$  for this part of the experiment?

$P(F|E) = \frac{5}{7}$ . Explain why.

What is the sample space  $S$  for this part of the experiment?

$$P(E \cap F) = \frac{6}{8} \times \frac{5}{7} = \frac{15}{28}.$$

# Product rule and tree diagrams

Probability tree diagrams are used with the product rule to find the probability for a sequence of events.

## Example 4

A box contains 6 good items and 2 defective items. Two items are selected at random from the box without replacement. Find the probability that both selected items are good.

Present the calculation in Example 3 on a **tree diagram**.

# Product rule for independent events

Events  $E$  and  $F$  are **independent events** if and only if

$$P(E \cap F) = P(E)P(F)$$

## Example 5

It is known that the failure of the hard drive of a computer is independent of the failure of the sound card. The probability of a hard drive failure on any given day is 0.00015, and the probability of a sound card failure 0.00025.

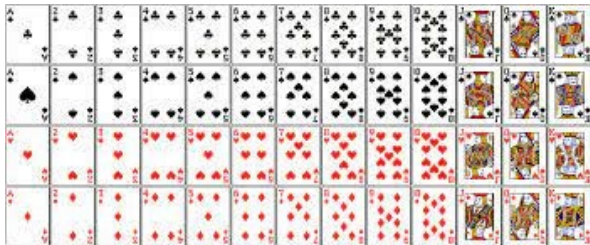
What is the probability on a given day of both a sound card and a hard drive failure?

**Solution**



# Exercises

1. Two cards are drawn from a standard deck, one after the other without replacement. Find the probability that
  - a) The second is a face card, given that the first card is a jack.
  - b) The second is an ace, given that the first card is not an ace.
  - c) Use the product rule to find the probability that the first card is a heart and the second is red.



# Monty Hall problem

On a television gameshow you are given the choice of three doors.

Behind one door is the Grand Prize; behind the others, Booby Prizes.

You pick a door, say Door A, and the host, who knows what is behind each door, opens another door, say Door B, revealing a Booby Prize.

The host then offers you the opportunity to change your selection to Door C. Should you stick with your original choice or switch?

Does it make any difference?

# Monty Hall simulation

```
N = 100000; # number of simulations
```

```
A = rep(0, N);
```

```
O = rep(0, N);
```

```
# Always choosing door A at the beginning
```

```
for(i in 1:N){
```

```
  x = sample(c(1,0,0), 3)  # Assigning the Grand Prize
```

```
  # Host action
```

```
  if(x[2] == 1) x = x[c(1,2)] # Removing the third door
```

```
  else x = x[c(1,3)]         # Removing the second door
```

```
  # Our choice
```

```
  if(x[1] == 1) A[i] = 1  # Staying with the first option
```

```
  else O[i] = 1          # Changing the option
```

```
}
```

```
mean(A) # Proportion of times we win staying with door A
```

```
mean(O) # Proportion of times we changing the door
```

Select Curriculum

IB DP Higher Level

Algebra

Functions & equations

Circular functions & trigonometry

Vectors

Statistics & probability

Calculus

Activities and tools

## Statistics & probability

5.1

5.2

5.3

5.4

5.5

5.6

5.7

5 Revision

5.4

### Conditional probability

Understanding how probabilities change according to some event which has taken place. Using tree diagrams for dependent events.

Lesson



Online homework



$$P(E|F) = \frac{P(E \cap F)}{P(F)} \text{ where } P(F) \neq 0.$$

## Turtles

- The average nest temperature determines the sex of baby sea turtles
- Embryos develop over 45-70 days
- The middle third of this incubation period is the temperature-sensitive period
- Average temperature above 31 degrees C: all female
- Average temperature below 27.7 degrees C: all male
- Pivotal temperature giving a 50% F/M ratio is 29 degrees.
  
- Write down some conditional probabilities

<https://www.seaturtlestatus.org/articles/how-temperature-determines-sex-in-sea-turtles>

## Turtles ctd

With respect to a certain nesting period in an given location:

Let H be the event that that the average temperature of a nesting period is at least 31 degrees.

Let C be the event that that the average temperature of a nesting period is less than or equal to 27.7 degrees.

Let M be the event that that the average temperature T of a nesting period is  $27.7 < T < 31$  degrees.

Are H, M and C mutually exclusive?

Find  $P(H)+P(M)+P(C)$ .

Could we work out the probability that we have all females in a given location and season? Do we need some extra information? Would these probabilities change from season to season?

# Space for a turtle tree diagram

# MATH503 Mathematical Concepts

## 2.1: Permutations



Probability

Multiplication principle

Factorial notation

Permutations

Formula for permutations

# Probability

Let  $S$  be a sample space of equally likely outcomes, and  $E$  an event.

The probability that event  $E$  occurs is  $P(E) = \frac{n(E)}{n(S)}$ .

When  $n(S)$  is large, it is not possible to list all the outcomes.  
Counting principles are needed to find  $n(S)$  and  $n(E)$ .

# Multiplication principle

Suppose  $n$  choices are to be made with

$m_1$  ways to make choice 1, and for each of these choices

$m_2$  ways to make choice 2, and so on

.....

$m_n$  ways to make choice  $n$ .

Then there are

$$m_1 \times m_2 \times \cdots \times m_n$$

different ways to make the sequence of choices.

# The multiplication principle

To find the number of choices, or arrangements of different types of item we multiply.

To find the total number of arrangements in probability trees we multiply the

|                                    |   |                                    |   |                                    |     |
|------------------------------------|---|------------------------------------|---|------------------------------------|-----|
| number of<br>choices<br>at event 1 | X | number of<br>choices<br>at event 2 | X | number of<br>choices<br>at event 3 | etc |
|------------------------------------|---|------------------------------------|---|------------------------------------|-----|

If we have choice of A things **and** then B things **and** then C things we **multiply**

If we have choice of A things **or** B things **or** C things we **add**

# Multiplication principle

## Example I

A combination lock can be set to open to any 3-letter sequence.

How many sequences are possible?

How many sequences are possible if no letter is repeated?

# Multiplication principle

## Example 2

A computer word is made up of 8 bits, each bit is a digit 0 or 1.

How many different computer words of this type is possible?

# Factorial notation

For any natural number (positive integer)  $n$ ,

$$n! = n \times (n - 1) \times (n - 2) \cdots \times 3 \times 2 \times 1 \text{ and } 0! = 1.$$

## Example 2

Simplify the following expressions:

►  $1! =$

►  $5! - 4! =$

►  $\frac{n!}{(n-2)!} =$

►  $\frac{(n-2)!}{(n-1)!} =$

►  $\frac{n!}{0!} =$

►  $n(n-1)! =$

►  $n! - (n-1)! =$

$$\frac{2!}{3!} =$$

$$\frac{3!}{2!3!} =$$

## In R

To calculate 5!

```
> factorial(5)
```

To calculate factorial of big numbers, we can always use logarithms, e.g.,  $\log(500!)$

```
> lfactorial(5)
```

To get more information, type in R

```
> ? factorial
```



# Permutations

A **permutation** of  $r$  elements from a set of  $n$  elements is any specific ordering of the  $r$  elements.

The number of permutations of  $n$  objects taken  $r$  at a time is written  $P(n, r)$ .

## Example 3:

In a club with 16 members, how many ways can a committee of 3 officers (president, secretary, treasurer) be chosen?

A set of 3 members have to be chosen from 16 members.

The order in which the 3 members are chosen is important.

Use the multiplication principle:

First choice (president):

Second choice (secretary):

Third choice (treasurer):

Number of ways to select the committee:

# Formula for permutations

General rule for  $0 \leq r \leq n$ :  $P(n, r) = \frac{n!}{(n-r)!}$

**Proof:**

**Example 4:**

Repeat example 3, but use a permutation to calculate the number of ways in which to select the committee

## Exercises

1. (a) How many 4-digit keys for an electronic lock can be made, if the first digit is a 1 or a 2, and no digits may be repeated?  
(b) How many 4-digit keys can be made, if the first digit is a 1 or a 2, and digits may be repeated?

# What did you get out of today?

What was the purpose of today's lecture?

What is one key thing you learnt?

Is there anything that puzzled me?